Nano-flowered manganese doped ferrite@PANI composite as energy storage electrode material for supercapacitors

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Abstract

Herein, we report a 3D nanoflower $MnFe_2O_4$@PANI composite as electrode material in supercapacitor. Manganese doped ferrites were synthesized by hydrothermal method followed by *in-situ* chemical oxidative polymerization of aniline to form a nanoflower composite. The nanoscale $MnFe_2O_4$ were infused with polyaniline (PANI) that offer superior synergistic effects by favouring access to charge/ion transit route, which is further improved by providing a broad surface area that is electrochemically active. It is entrenched that $MnFe_2O_4$@PANI involves charge transfer pseudocapacitance at both the ionic sites of Mn and Fe and oxidized emeraldine salt of PANI, balanced by the inclusion/exclusion of proton within/outside the lattice. On the one hand, the fluffy nature of PANI provided channels for charge transport, and manganese ferrites helped in storing the electrons through changes in the valence state of the active sites. We obtained a high specific capacitance of 623 F/g at a current density of 1 A/g. A prototype of the developed supercapacitor showed excellent device performance with a high energy density of 179 Wh/kg and maximum power density of 982 W/kg with ~95% retention of specific capacitance after 10,000 cycles.

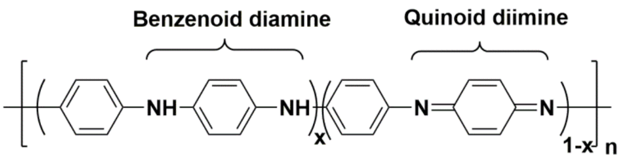
1. Introduction

Storing electrical charge has been known since ancient times through rubbing of amber with a piece of cloth which would attract small hairs or another light object. The essential matter of content in deciding the cost and economics of energy storage is the structure of electricity markets and the method through which they are operated and regulated. Currently, there are eight most promising and emerging energy storage technologies viz. compressed air energy systems, flywheels, flow batteries, superconducting magnets, pumped hydropower, sodium-sulphur batteries, lithium-ion batteries and supercapacitors [1,2,3].

Among these, batteries and supercapacitors are predominantly used energy storage systems. Batteries have a history of more than 150 years; they provide excellent output performance over a broad range of temperature, assuring high reliability, but are inefficient for long-term use (>10 years). Additionally, they might cause safety and environmental issues because of the involvement of various chemicals during fabrication. Extensive use and need for a new energy storage technology led to the development of a next-generation energy storage device, i.e. supercapacitor (SC). SCs have the ability to be fully charged and discharged in seconds because of their energy storage mechanism and therefore has the edge over other energy devices like batteries etc. SC stores energy by charge separation as ions which are bind to the electrodes to store energy. The escalation in energy with an increase in surface area of electrode motivated researchers to focus on the development of electrodes that exhibits high surface area to procure an SC with high capacitance value [1,4,5].

Based on the energy storage mechanism, SCs are classified into an electrochemical double layer capacitor (EDLC) and pseudocapacitor. EDLC reserve charge by adsorption of electrolyte ions at the interface of electrode-electrolyte, while pseudocapacitors reserve charge by faradic reactions at the surface of the electrode [6,7]. Generally, carbon-based materials (like CNTs) are used as electrode material in EDLC due to its conductivity and high surface area; however, they also acquire high synthesis charges and high contact resistance within the electrode and current collector [8,9]. Pseudocapacitors, on the other hand, use transition metal oxides and conducting polymer-based materials, exhibiting large specific capacitance than EDLCs. These materials use highly reversible surface redox reactions to store electrochemical charge. Ruthenium oxides are used extensively for its high specific capacitance; however, it is difficult to commercialize because of its rare availability, high cost and toxic nature. This certainly has restricted RuO_2 processing on a vast scale production. However, the drawbacks have led to the emergence of different metal oxides (such as MnO_2, Fe_3O_4, NiO, ZnO_2, Co_3O_4, etc.) for use as active electrode material in energy devices [10–12,13]. Bimetallic oxides and metal ferrites ($M_xFe_2O_4$, where M = Mn, Zn, Co, Ni, etc.) are also considered as a promising candidate due to their higher electric conductivity and electrochemical activity due to the presence of multiple valence cations [14]. Among them, manganese ferrite ($MnFe_2O_4$) nanoparticles have gained much attention because of their chemical endurance and high capability in bountiful magnetic applications, its use as an anode material in Li-ion battery, biosensors and drug delivery applications. However, the restricted conductivity of $MnFe_2O_4$ and other pseudocapacitive metal oxides induce limited specific power and high electrode resistance compared to other forms of SC [15,16,17]. It is also envisaged that coating a conducting material with electrochemically active species can improve the transportation of charges [18].

Conducting polymers like polyaniline and polypyrrole have fascinated great interest in energy storage devices, sensors and electrochromic devices; because of its high conductivity, high capacitive property and low-cost [19]. Nanocomposites, combining metal oxide nanoparticles with a conducting polymer, and/or carbonaceous material has been found to be instrumental in improving the power capacity and rate capacitance of energy devices [20]. The emeraldine salt (ES) of polyaniline is more promising due to good environmental stability, multi-redox reactions, low-cost for its infinite abundance, and improved electrochemical performances. PANI exists in three forms based on their oxidation states *viz.* leucoemeraldine (LE), emeraldine (E), and pernigraniline (PNA) (**Scheme 1**) [21]. PANI can be prepared easily *via.* chemical or electrochemical method using acidic media, resulting in the form of powder or thin film. Different composites with different ratios have already been utilized successfully such as $MnFe_2O_4$/Graphene/PANI[17], DAS-RGO/PANI [22], CS/PANI [23], $MnFe_2O_4$/Carbon black/PANI [24], N-doped porous carbon/PANI [20].



Scheme 1. Schematic presentation of oxidation states of PANI, n is the polymerization degree, x=1 for leucoemeraldine form, x=0.5 for emeraldine form, and x=0 for pernigraniline form.

Loche et al. 2018 studied the dielectric ion-surface interaction between graphene layer and hydrated ion; while Ouyang et al. 2019 reviewed various strategies investigated by the researchers in order to understand the behaviour of charge carriers in metal oxides and improve their performances [25,26]. In contrast to the capacitive performance of the material, there is only a handful of literature on the role of morphology and working mechanism of the pseudocapacitive material [27].

The present work aims at the study of synergistic charge storage reaction mechanism in the $MnFe_2O_4$@PANI for building an ultra-high rate pseudocapacitor. Herein, the electrochemical properties, interaction and charge storage mechanism in $MnFe_2O_4$@PANI has been studied in detail using X-ray photon and impedance spectroscopy. We have reported a simple route for preparing low-cost $MnFe_2O_4$ nanoparticles in a green solvent, which is combined with aniline to form $MnFe_2O_4$@PANI nanoflowers. In order to avoid the agglomeration between the manganese ferrite nanoparticles, we have carried out *in-situ* chemical oxidative polymerization of aniline to obtain $MnFe_2O_4$@PANI nanoflowers. The as-prepared nanoflowers were synthesized with different ratios under the similar quantity of aniline precursor, and the resulting composite were characterized electrochemically. The effects of concentration of $MnFe_2O_4$ nanoparticles on the electrochemical properties of $MnFe_2O_4$@PANI nanoflowers were examined by galvanostatic charge-discharge method up to

10,000 cycles. The symmetric supercapacitor showed energy density as high as of 179 Wh kg^{-1} and maximum power density of 982 W kg^{-1} with a high specific capacitance of 623 F/g in an aqueous solution of 1M H_2SO_4 with the working voltage of 1.1 V. **Table S1** summarizes a comparison of $MnFe_2O_4$@PANI with other reported materials on the basis of their performance on the supercapacitor devices [28–30]. The device can retain specific capacitance of ~95% even after 10,000 charge-discharge cycles at a high rate.

2. Experimental methods

2.1.1. Synthesis of $MnFe_2O_4$ Nanoparticles

$MnFe_2O_4$ nano-ferrites were prepared by hydrothermal method. Firstly a suspension of metal oxalates was prepared by taking a stoichiometric amount of 1 mmol of iron (III) chloride hexahydrate $FeCl_3.6H_2O$, and 0.5 mmol of manganese (II) chloride $MnCl_2$ in deionized water and sonicated. Oxalic acid dihydrate $C_2H_6O_6$ was then added as a chelating precipitant. The molar ratio of metal salts to oxalic acid was 0.5:1:2.6 for salts of Mn^{+2}, Fe^{+3} and oxalic acid, respectively. To this solution, 0.4 ml of 20% aqueous tetra n-butyl ammonium bromide was added for peptization. To deprotonate oxalic acid and induce co-precipitation of metal oxalates, the pH of the metal oxalate solution was raised to pH 12 with 5M sodium hydroxide NaOH at room temperature. The mixture was stirred mechanically while adding sodium hydroxide. Later, the solution mixture was transferred in an autoclave reactor bomb and placed in a preheated oven at 145 ˚C for 24 hours. The mixture was allowed to cool down to room temperature, which resulted in the blackish-brown precipitate of manganese ferrite. The precipitate was washed with distilled water several times and purified magnetically. The precipitate was dried overnight in a hot air oven at 60 ˚C. The present method was accomplished through a simple process without using any organic solvents, and hence it was possible to mass-produce uniform magnetic manganese-ferrites.

2.1.2. Synthesis of 10%$MnFe_2O_4$@PANI Nanoflowers

10%$MnFe_2O_4$@PANI nanoflowers were prepared using *in-situ* chemical oxidative polymerization method, where $MnFe_2O_4$ nanoparticles were allowed to stir for 1 hour to obtain a uniform suspension. Then, 2.5 ml of aniline was added slowly in a 500 ml round bottom flask followed by stirring in the ice bath for 1 hour. Further 1M HCl was added (100 ml). After complete dissolution of aniline, 0.1M ammonium persulphate [$(NH_4)_2S_2O_8$] solution was added dropwise in such a way that the addition is completed in 30 minutes, maintaining the temperature below 5 °C. The reaction was allowed to continue for 4 hours at the end of the reaction, nanocomposites were filtered through sintered glass crucible and washed with distilled water until a colourless filtrate is obtained. The sample was then collected in a Petri dish and is dried overnight in a hot air oven at 60 °C. PANI coated $MnFe_2O_4$ nanoflowers was prepared using varying stoichiometry of nanoparticle (w/w 2.5%, 5%, 20% and 50%) w.r.t. aniline monomer.

2.2. Characterization

The composition, morphology, and structure of the as-prepared nano-ferrites, polymer and the nanoflower composites were characterized. X-ray diffraction (XRD) was performed to analyze the phase structure and crystalline nature of the ferrites and the composites. The surface morphology was examined under Field emission scanning electron microscope (FE-SEM) and High-resolution transmission electron microscope (HR-TEM). Raman spectroscopy was used to investigate the Raman vibrations of the as-prepared samples. X-ray photoelectron spectroscopy (XPS) measurements were performed to further confirm the elemental composition of the samples. The electrochemical properties were studied using Autolab PGSTAT204 potentiostat for CV and EIS analysis, whereas, Arbin battery testing instrument (BT2000, USA) was used for GCD analysis of the $MnFe_2O_4$@PANI composite electrode.

2.3. Electrochemical sample preparation

2.3.1. Assembly and electrode preparation for three-electrode system

Three electrode systems consist of a counter electrode, reference electrode and a working electrode. A high surface area platinum wire is used as a counter electrode, Ag/AgCl as a reference electrode and glassy carbon on which the electroactive materials were drop-casted was used as a working electrode. 100 µg of $MnFe_2O_4$ nanoferrite electrode materials that was dispersed in iso-propyl alcohol was casted on the working electrode by drop-cast method. The electrodes were allowed to dry at room temperature for electrochemical studies.

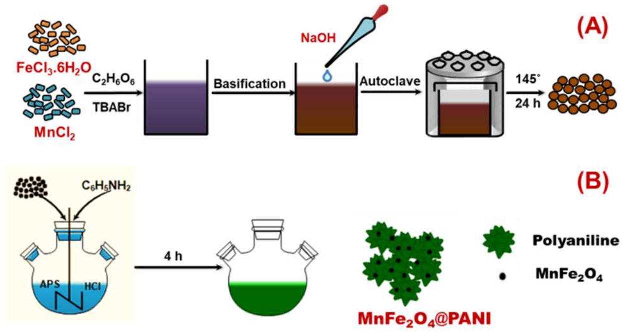
2.3.2. Assembly and electrode preparation for two-electrode system

The electrodes were fabricated by mixing the active material ($MnFe_2O_4$@PANI), binder PTFE (poly-tetrafluoroethylene) and carbon black in D/W. Fig. S4(S1) illustrates the schematic representation for fabrication of a two-electrode system for evaluating its potential application in supercapacitor. Firstly, dough was prepared by mixing the 80 wt% active material (10%$MnFe_2O_4$@PANI composite), 15 wt% of carbon (carbon black, C65) and 5 wt% of PTFE in 8-10 drops of distilled water with frequent and gentle kneading until it reached plasticity. The dough was rolled instantaneously to obtain a thin film of the electrode and the residual solvent was removed by drying overnight in a hot-air oven.

The electrode film was punched to a 10 mm diameter disc-shape. The weights of the active electrode materials used for the electrochemical estimations were ~2.2 mg cm^{-2}. The electrodes were loaded into a two-electrode swagelok-cell using cellulose filter paper as a separator for electrochemical testing. Fig. S4(S1) illustrates the arrangement of electrodes and separator in the swagelok cell. To ensure the penetration of electrolyte into the electrodes and separator, the as-prepared electrodes were aged for an hour before any electrochemical testing.

3. Results and discussion

The performance of the energy storage device is primarily determined by the electrode material, which is crucial in enhancing the pseudocapacitance of the device. Astounding progress has been seen in chemical engineering in developing new electrode materials to suffice the requirement of energy devices. With respect to the synthetic method of metal oxides/polymer nanocomposites, several not only hazardous chemicals and organic solvents are used but are also difficult to scale up for large-scale production of the nanomaterial. Herein, we report a simple and greener synthetic method for metal oxides/polymer nanocomposite, which can be scaled up for commercial application. **Scheme 2** depicts the pictorial presentation of the novel fabrication process of $MnFe_2O_4$@PANI nanoflowers; wherein, water is used as a solvent proceeding with low-cost precursor metal salts to obtain a good yield of the nanoflower composites.



Scheme 2. Schematic representation of the synthesis of $MnFe_2O_4$@PANI nanoflowers, (A) Illustrates the formation of $MnFe_2O_4$ nanoparticles using the hydrothermal method, and (B) Illustrates the in-situ chemical oxidative polymerization to form PANI nanoflower in the presence of $MnFe_2O_4$ NP's to form $MnFe_2O_4$@PANI nanoflowers.

Herein, ammonia released from ammonium persulphate reduces the interfacial energy and smoothens the composite surfaces by Ostwald's ripening phenomenon that creates nanoflower formation [31]. Five different compositions were prepared for the composites *viz.* 2.5%, 5%, 10%, 20%, and 50% $MnFe_2O_4$@PANI with the ratio of $MnFe_2O_4$:PANI as 2.5:97.5, 5:95, 10:90, 20:80 and 50:50 respectively; out of which 10% $MnFe_2O_4$@PANI exhibited superior electrochemical performance as compared to the other compositions (discussed in the later part **Fig. 5a, 5b**).

3.1. Morphology and structural analysis

The surface morphology, shape and size of $MnFe_2O_4$, PANI and $MnFe_2O_4$@PANI nanoflower were analyzed by FE-SEM (**Fig.S1**). Nano-ferrite particles were polygonal-shaped; whereas, PANI had uniform flower-shaped flaky nanostructure which has an anatomy of a marigold flower (**Fig. S1b**). **Fig. 1a** shows SEM image of 10%$MnFe_2O_4$@PANI nanoflower. The purity and elemental composition of the composites were estimated by elemental mapping (**Fig. S1**) and EDS spectra (**Fig. S2**). The moderate attractive force, i.e. the hydrogen bond formed between the nano-ferrites and polymer (discussed in the later part) held the 10%$MnFe_2O_4$@PANI nanoflower composites together, and the repulsive forces between the nanoparticles limit the aggregation of the crystalline $MnFe_2O_4$ [32]. As per the ratio of $MnFe_2O_4$ w.r.t. Polyaniline, which is 10:90 respectively, the nanosized particles of $MnFe_2O_4$ were found to be distributed on the surface of polyaniline in the bulk of nanoflower composites.

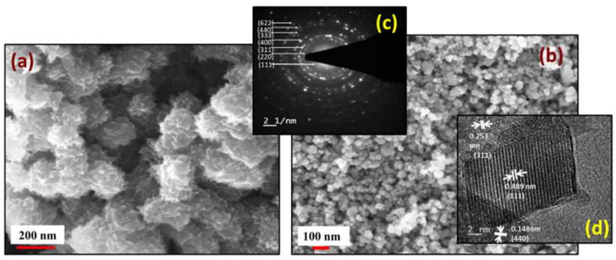


Fig. 1 FE-SEM image of (a) 10%$MnFe_2O_4$@PANI nanoflower, (b) $MnFe_2O_4$, (c) SAED and (d) HRTEM image of $MnFe_2O_4$ nanoparticles.

The HR-TEM image showed the crystallinity of $MnFe_2O_4$ nanoparticles. A group of atomic planes corresponded to the lattice margins within the single crystal particle. The lattice fringes with inter-planar distances of 2.53 A°, 4.89 A° and 1.48 A° were consistent with the (311), (111) and (440) planes respectively of cubic $MnFe_2O_4$ (**Fig. 1d**). The selected area electron diffraction (SAED) pattern (**Fig. 1c**) showed that the particles are crystalline and can be indexed to a cubic spinel structure as also evident from XRD. The SAED pattern indicated bright diffraction spots and set of rings, corresponding to the reflections from seven crystal planes, viz., (111), (220), (311), (400), (333), (440) and (622). This validated the formation of the spinel cubic phase of the $MnFe_2O_4$ nanoparticles.

Primary characterization for the interaction between the conducting polymer and $MnFe_2O_4$ nanoparticle in the 10%$MnFe_2O_4$@PANI composites was recorded using FT-IR spectra (**Fig. 2a**) on JASCO FT/IR – 460 Plus. The vibrational spectra of $MnFe_2O_4$ (spinel ferrites) attribute to the tetrahedral sites, *i.e.* the high-frequency intrinsic vibration (600–580 cm^{-1}) and the octahedral sites, *i.e.* the low-frequency band (440–400 cm^{-1}).

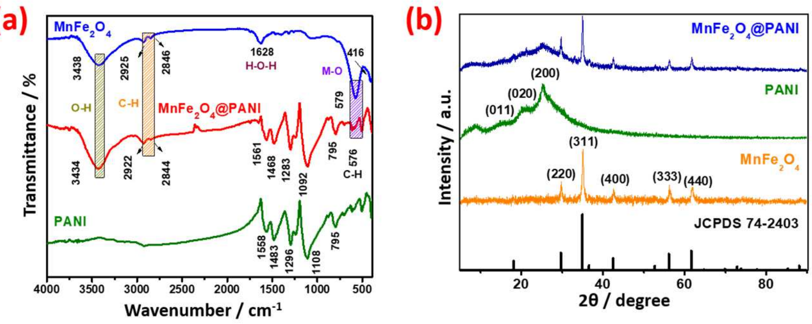


Fig. 2 (a) FT-IR spectra of PANI coated $MnFe_2O_4$ nanoflower composites, (b) XRD patterns of $MnFe_2O_4$, PANI and 10%$MnFe_2O_4$@PANI nanoflower.

The absorption peaks at 579 cm^{-1} and 416 cm^{-1} are the characteristic peak of Fe-O and Mn-O bond respectively, the mode of vibration provided spinel ferrite structure formation, as evidence for the formation of the metal ions-oxygen bond in the tetrahedral and octahedral sites [33,34]. The FTIR spectrum of 10%$MnFe_2O_4$@PANI showed a peak at 3434 cm^{-1} which is attributed to the stretching vibration of the hydroxyl group on the nanoparticle surface. Peaks at 2925 cm^{-1} and 2846 cm^{-1} are due to alkyl chain stretching (C–H stretch) of physically adsorbed tetra-N-butyl ammonium bromide on manganese ferrite surface. Peak at 1628 cm^{-1} is attributed to hydrogen bending vibrations of water (H-O-H). Prominent peaks at 1558 cm^{-1} for PANI and 1561 cm^{-1} for 10%$MnFe_2O_4$@PANI corresponds to benzenic-quinonic nitrogen, while bands at 1483 cm^{-1} for PANI and 1468 cm^{-1} for 10% $MnFe_2O_4$@PANI corresponds to C-C aromatic stretch [35]. Peaks at 795, 1092 and ~1296 cm^{-1} in 10%$MnFe_2O_4$@PANI represents C-H stretch out-of-plane, polyconjugated system and the C-N-C stretching respectively [36,37]. Thus, the presence of reproducible peaks in 10%$MnFe_2O_4$@PANI at different wavenumbers is indicative of the formation of manganese ferrite@polyaniline nanocomposites.

X-ray diffraction was performed to learn the phase and crystalline nature of 10%$MnFe_2O_4$@PANI nanocomposites w.r.t. individually prepared polyaniline (PANI) and $MnFe_2O_4$ nanoparticle (**Fig. 2b**). The XRD patterns of as-prepared $MnFe_2O_4$ nanoparticle matched with the standard data (JCPDS card No. 74-2403) which showed the main diffraction patterns indexed for cubic spinel. The spectra for $MnFe_2O_4$ did not show any other peaks for impurities. Debye–Scherrer formula, to estimate the average particle size from FWHM (full width at half maximum), were studied:

$$D = 0.89\lambda / \beta cos\theta$$

where D is mean crystallite size (in nm), λ refers to Cu Kα wavelength (0.154 nm), β is FWHM of diffraction peaks, and θ is Bragg's angle [38]. The average crystallite size of $MnFe_2O_4$ is examined to be ~18 nm. The XRD pattern of PANI exhibited amorphous nature in the partially crystalline state with three diffraction peaks of 2θ at about 14.99˚, 20.3˚ and 25.21˚. The benzenoid and quinoid ring planes of PANI are responsible for crystalline structure [23,39]. The XRD patterns of $MnFe_2O_4$

and PANI were used to match diffracted crystal planes of 10%$MnFe_2O_4$@PANI nanocomposites, which showed a combination of both diffraction peaks obtained from free PANI and $MnFe_2O_4$, ensuring the formation of the nanocomposite. This refers to the partial distortion of $MnFe_2O_4$ crystal structure during the polymerization reaction.

Furthermore, Fig. S12 shows the nitrogen adsorption–desorption isotherms of the synthesised electrode materials to present its specific textural properties. In terms of shape, according to the IUPAC classification the isotherm can be classified as type IV. No significant delay in the capillary evaporation with respect to the capillary condensation of nitrogen and the porous structure is quite open as indicated by the narrow hysteresis loop at a low relative pressure. The BET surface area of $MnFe_2O_4$, PANI and $MnFe_2O_4$@PANI are 59.6 m^2/g, 22.2 m^2/g and 79.6 m^2/g respectively. The surface area follows the order as: $MnFe_2O_4$@PANI > $MnFe_2O_4$ > PANI. The extra surface area of $MnFe_2O_4$@PANI nanocomposite is derived from the $MnFe_2O_4$ nanoferrite and the flaky structure of PANI.

3.2. Surface analysis

To investigate the mechanism of interaction with surface atoms in the $MnFe_2O_4$@PANI nanoflower, XPS was carried out. The results showed the presence of both Fe and Mn ions in 10%$MnFe_2O_4$@PANI nanoflower composites (**Fig. 3e**). The survey spectrum of $MnFe_2O_4$ (**Fig. 3c**) shows the presence of the elements Mn, Fe and O. Furthermore, the atomic concentrations of Fe*2p* and Mn*2p* in $MnFe_2O_4$ were found to be 19.95% and 8.76%, respectively, which revealed the molar ratio of Mn:Fe close to 1:2. **Fig. S3** illustrates the high-resolution XPS core-level spectra of O*1s*, Mn*2p*, and Fe*2p* of $MnFe_2O_4$ nanoparticles, where Mn2*p* spectrum shows Mn in +2 and +3 oxidation states (**Fig. S3b**). The core-level emission peaks of Mn*2p*, Fe*2p* and O*1s* are given in detail in **Table S2**. The peak $Fe^{2+}2p_{3/2}$ and peak $Fe^{3+}2p_{3/2}$ at 710.34 eV and at 711.69 eV were paired with a satellite peak at 12.2 eV and 11.39 eV respectively, which was observed above the main peak. This showed the existence of Fe^{3+} and Fe^{2+} oxidation states in the $MnFe_2O_4$ nanoparticles [40]. The core-level spectra of O1*s* (**Fig. 3f**) corresponds to two distinct oxygen species and one satellite peak. The low binding energy at 530.0 eV along with a satellite peak at 11.83 eV can be attributed to Fe-O and Mn-O bonds. The high binding energy at 531.23 eV, referred to oxygen forming a double bond to iron (O-Fe=O) [41].

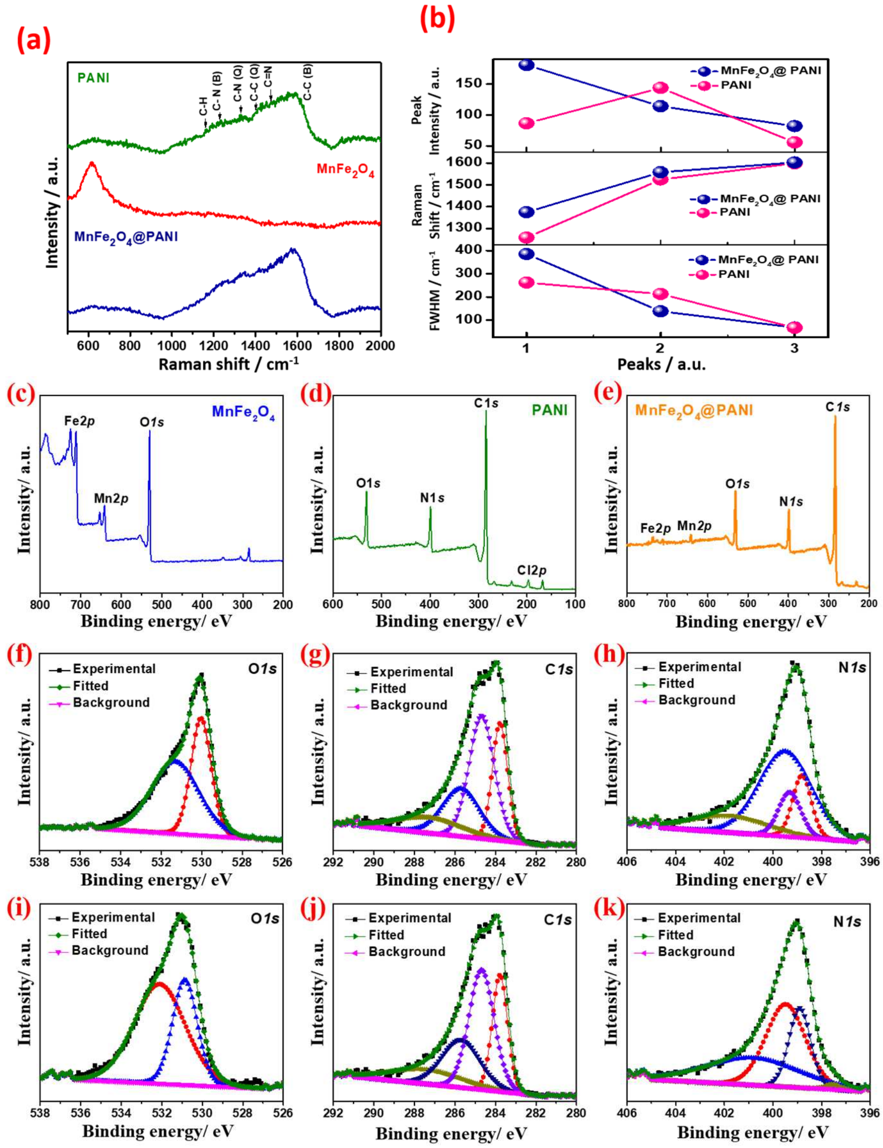


Fig. 3 (a) Raman spectra of PANI, $MnFe_2O_4$, and 10%$MnFe_2O_4$@PANI nanoflower, (b) Fit parameters from Raman spectra for Polyaniline and $MnFe_2O_4$@PANI. The distinguishable band peaks, as indicated for the samples, are fitted using Gaussian profile. XPS wide scan of as-prepared (c) $MnFe_2O_4$, (d) PANI, and (e) 10%$MnFe_2O_4$@PANI nanoflower. Core level XPS spectra of as-prepared (f) O*1s* spectra ($MnFe_2O_4$), (g) C*1s* spectra (PANI), (h) N*1s* spectra (PANI), (i) O*1s*, (j) C*1s*, and (k) N*1s* spectra of 10%$MnFe_2O_4$@PANI nanoflower.

The XPS scan of PANI (**Fig. 3d**) showed the presence of nitrogen (N*1s* ~399 eV), oxygen (O*1s* ~532 eV), carbon (C*1s* ~284 eV), and chlorine (Cl*2p* ~197 eV). Oxygen was derived from the oxidation of the film's surface partially, of the emeraldine salt form of PANI [42]. Elements, nitrogen and carbon originated from the polymer backbone, whereas chlorine being the counter ion. The core-level spectra of N*1s* (**Fig. 3h**) were deconvoluted into four peaks complementary to the benzenoid/di-amine nitrogen and quinoid/di-imine centred at 399.39 eV and 398.9 eV, respectively. Peaks at 402.79 eV and 400.93 are due to protonated imine and positively charged nitrogen, the oxidized amine, respectively [19]. Electronic defects have been produced by PANI in polymer chains (polarons/bipolarons) on protonation, which is responsible for charge transport [43]. The core-level spectra of C*1s* (**Fig. 3g**) were deconvoluted into four peaks with extended binding energies ranging from ~289 eV to 284 eV and are mentioned in detail in **Table S2**. A broad peak at ~285 eV represents an asymmetric broadening towards higher binding energy because of a mixture of protonated amine and imine sites (also called a shake-up process). Second binding energy tail ~ 289 eV was due to the π–π^*. In the spectrum of 10%$MnFe_2O_4$@PANI composite (**Fig. 3e**), the elements Mn, Fe, C, N, and O could be observed. Based on the above observations, the interaction of nano-ferrites with polyaniline can be elucidated.

3.3. Proposed mechanism obtained from Raman spectra and XPS

Raman spectra of $MnFe_2O_4$, PANI, and 10%$MnFe_2O_4$@PANI were recorded from 500 to 2000 cm^{-1} (**Fig. 3a**). PANI exhibited emeraldine pattern with the prominent functional groups found at ~1597 cm^{-1} (C-C stretch, benzenoid ring), ~1555 cm^{-1} (C=C stretch, quinoid ring), ~1481 cm^{-1} (C=N stretch, quinoid ring), ~1406 cm^{-1} (C-C stretch, quinoid ring), ~1329 cm^{-1} (C-N stretch, quinoid ring), ~1233 cm^{-1} (C-N stretch, benzenoid ring), and ~1166 cm^{-1} (C-H bending, quinoid ring) [44]. The vibrational modes of $MnFe_2O_4$ were observed at 613 cm^{-1} [21]. The 10%$MnFe_2O_4$@PANI nanocomposites exhibited a shift in the peaks of PANI. However, the Raman signal of $MnFe_2O_4$ was not clear in the nanocomposite, probably due to a higher wt% ratio of PANI over $MnFe_2O_4$ (PANI: $MnFe_2O_4$; 90:10). Using Gaussian shape the intensity, FWHM and peak position has been determined by profile fitting of prominent bands (**Fig. S11**). Nature of the deviation of peak parameters (**Fig. 3b**) was summarized as follows: (1) Raman peak width and peak intensities for 10%$MnFe_2O_4$@PANI showed an increasing trend for first and third peak while the second peak showed decreasing trend w.r.t. PANI, (2) the intensity of the centre peak (Raman shift) of all the bands of 10%$MnFe_2O_4$@PANI increased with the shift in band positions to higher wavenumbers significantly. Hence, the shift in peak and change of chemical environment indicated the successful formation of 10%$MnFe_2O_4$@PANI. Furthermore, as the nanocomposites gain more energy, band shift can be observed at higher wavenumbers, revealing the significance of blue shift, which is due to the chemical interactions between the nano-ferrites and conducting polymers.

By comparison, the O*1s* spectrum of $MnFe_2O_4$@PANI nanoflower (**Table S4**), showed a peak shift to 530. 85 and 532.206 eV, where, this offset of 0.79 and 1.009 eV can be presumed to be an outcome of the interaction between the N-H (hydrogen) of polyaniline and O (oxygen) of $MnFe_2O_4$. This is further compared and confirmed in the N*1s* spectrum of $MnFe_2O_4$@PANI nanoflower (**Table S5**), where the peak shifts to 401.233 and 403.135 eV, with an offset of 0.337 and 0.645 eV. **Fig. 4** illustrates the mechanistic interaction between $MnFe_2O_4$ and PANI in $MnFe_2O_4$@PANI nanoflower (obtained results confirm the formation of the hydrogen bond; theoretically, 2-40 kcal/mol of energy is required for hydrogen bond formation) [45]. Lastly, overall 20 kcal/mol (i.e. 0.86 eV) energy is required for electrostatic attraction which is achieved by the composite giving an increase in the shift by 0.843 eV.

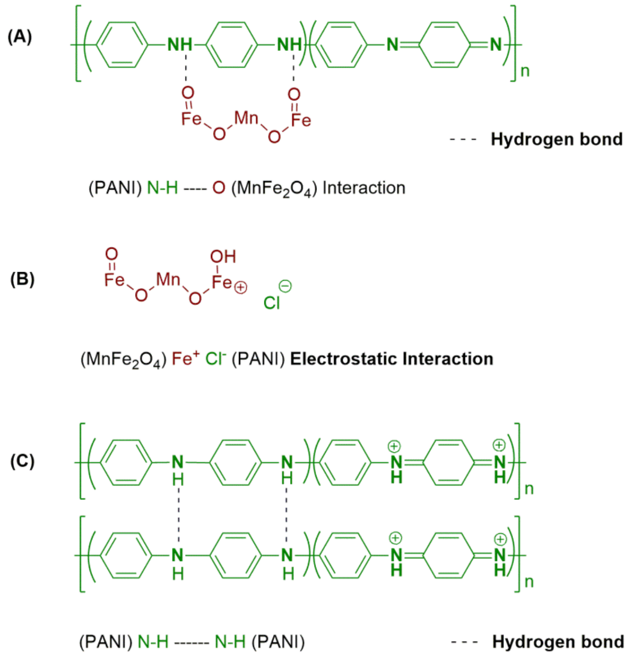


Fig. 4 Illustrates the possible mechanistic interaction of $MnFe_2O_4$ onto PANI in the 10%$MnFe_2O_4$@PANI nanoflowers based on the XPS shift, (A) H-bonding between PANI and ferrite, (B) Electrostatic interaction between metal ion of ferrite and chloride ions of polyaniline, and (C) a strong H-bond formation among polyaniline chains

3.4. Electrode fabrication

Fig. S4 illustrates the schematic representation of the fabrication of a two-electrode system for use in supercapacitor. Firstly, a dough is prepared by mixing 80 wt% of active material (10%$MnFe_2O_4$@PANI nanocomposite), 15 wt% of carbon (carbon black, C65) and 5 wt% of PTFE (binder) in 8-10 drops of distilled water with frequent and gentle kneading until the dough reaches plasticity. Later, the dough was rolled immediately to obtain a thin film of the electrode, and the residual solvent is removed by drying overnight in a hot-air oven.

3.4.1. Electrochemical performances

The electrochemical behaviour of 10%$MnFe_2O_4$@PANI nanoflowers was investigated by fabricating a two-electrode system swagelok cell prototype to observe the electrochemical mechanism of the metal ferrite@conducting polymer composite. However, the electrochemical behaviour of all the systems was assessed in a 3-electrode system before fabricating the 2-electrode prototype.

The electrochemical performance of $MnFe_2O_4$ nanoparticles, polyaniline and different compositions of nanocomposites *viz.* 2.5%$MnFe_2O_4$@PANI, 5%$MnFe_2O_4$@PANI, 10%$MnFe_2O_4$@PANI, 20%$MnFe_2O_4$@PANI, and 50%$MnFe_2O_4$@PANI electrodes were examined by cyclic voltammetry (CV) in the potential window of -0.1 to +1.0 V vs. Ag/AgCl at a scan rate of 50mV/s in 1M H_2SO_4 to see the redox behaviour of the electrode (**Fig. 5b**). **Fig. 5a** illustrates the plot of PANI and different compositions of the nanocomposites against specific capacitance. The conductive network and electrochemical properties of PANI were found to improve on the addition of $MnFe_2O_4$ nanoparticles. This may be due to the interaction between the polymer chain and $MnFe_2O_4$ nanoparticles which led to an increase in conducting network by the formation of H-bonds. As can be seen from **Fig. 5a**, 10%$MnFe_2O_4$@PANI nanoflower gave the highest specific capacitance, followed by 20%$MnFe_2O_4$@PANI, and 50%$MnFe_2O_4$@PANI. Integrating **Fig. S5c** and **5b** the non-rectangular curves were obtained for the nanoflowers which did not show constant capacitance throughout the potential window [46–48]. The steady decrease in specific capacitance of the composites with an increase in the concentrations of nanoparticle (>10 wt%) can be attributed to the obstructions in the conducting path created by the nanoparticles entrenched in the PANI matrix**. Fig. S5 and Fig. 5b** depicts the cyclic voltammogram of polyaniline and all five nanocomposites of different wt% ratios. At the same scan rate of 50 mV/s (**Fig. S10**), the capacity of 10%$MnFe_2O_4$@PANI was calculated using the equation given below,

$C_s = \int Idv/mv3.6$ *(1)*

where $\int Idv$ is the area under the CV curve, v is scan rate, and m is the mass of the electrode. The capacity was found to be 900 mAh/g. PANI exhibited three redox pairs (**Fig. S5b**) of their transition from one form to another. The conventional redox pairs were obtained at 0.2/0.13 V, which can be attributed to the conversion of leucoemeraldine to emeraldine states. The peaks 0.5/0.46 V can be assigned to the exchange of emeraldine to pernigraniline [27]. The redox pair at 0.76/0.73 V corresponds to another transition from pernigraniline to leucoemeraldine on accepting the protons from the acidic electrolyte. On the other hand, hardly any peak was visible for $MnFe_2O_4$ as compared to the other two because of insignificant capacitance (**Fig. S5d**).

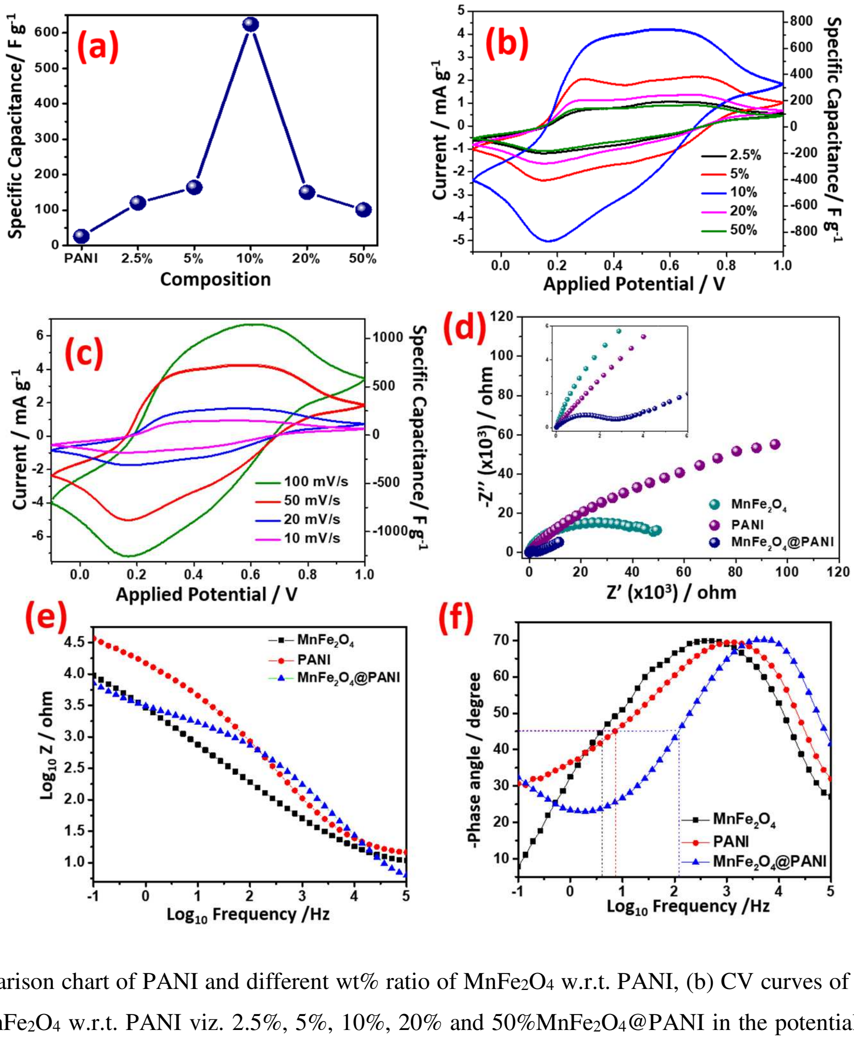


Fig. 5 (a) Comparison chart of PANI and different wt% ratio of $MnFe_2O_4$ w.r.t. PANI, (b) CV curves of PANI and different wt% ratio of $MnFe_2O_4$ w.r.t. PANI viz. 2.5%, 5%, 10%, 20% and 50%$MnFe_2O_4$@PANI in the potential window of -0.1 to +1.0 V vs. Ag/AgCl at a scan rate of 50mV/s in 1M H_2SO_4, (c) CV profile of 10%$MnFe_2O_4$@PANI electrode in the potential window of -0.1 to +1.0 V vs. Ag/AgCl in 1M H_2SO_4 at different scan rate, (d) Illustrates electrochemical impedance spectra (Nyquist plot) of $MnFe_2O_4$, PANI and 10%$MnFe_2O_4$@PANI nanoflowers. (Inset, magnified part of the high-frequency

region), (e) Bode plot of Impedance against Frequency, and (f) Bode plot of Phase angle against Frequency in 1M H_2SO_4 aqueous electrolyte from 100 kHz to 0.01 Hz with an amplitude of 50mV.

Unlike PANI, the 10%$MnFe_2O_4$@PANI nanoflowers exhibited an increase in specific capacitance after addition of $MnFe_2O_4$. The synergistic effect displays a significant advantage during the electrochemical process of the nanoflower electrode. The synergistic effect was probably due to the PANI, whereas $MnFe_2O_4$ provides excellent electrochemical activity due to the presence of multiple valence states as compared to other metal oxides [49]. The nano-ferrites provide higher surface area for storing the charges; they also provided approachable sites for conduction of electrons. The transition from Fe^{2+} to Fe^{3+} and Mn^{3+} to Mn^{2+} led to the movement of electrons and the generation of holes. Nevertheless, the steady decrease in specific capacitance with increasing concentration of nanoparticles was due to the increased particle-size formed due to increased addition of $MnFe_2O_4$ particles which might offset enhance in specific capacitance from the redox process. The electron transfer mechanism with the electrolyte is illustrated in **Fig. 6**. The electrolyte supplied protons to the polymer which has several channels for transporting charges, while, it gets stored in the nano-ferrites.

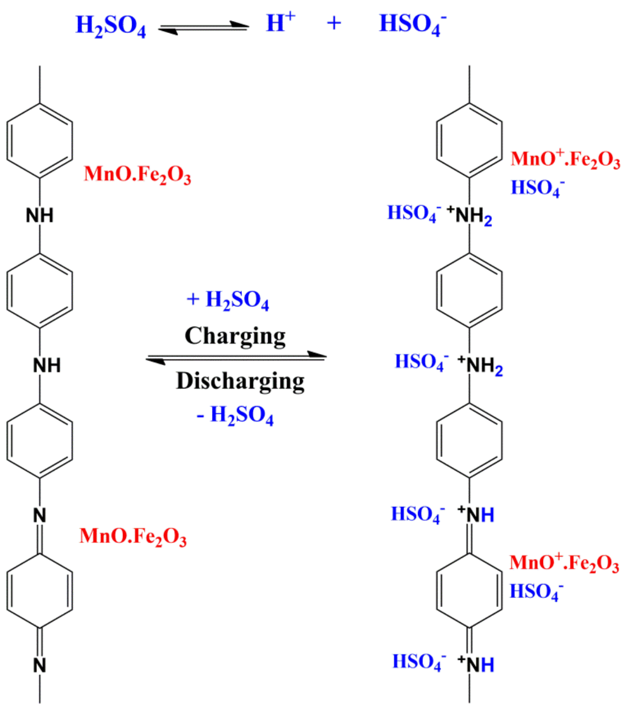


Fig. 6 Illustrates the proposed mechanism while charging and discharging

The area under the curve (AUC) of 10%$MnFe_2O_4$@PANI was found to be more than the AUC of $MnFe_2O_4$, and PANI (**Fig. S5d**) which is visible in the charge-discharge curves performed at a current density of 1 A/g in 1.1 V of voltage window of all the systems viz., $MnFe_2O_4$, PANI and $MnFe_2O_4$@PANI (**Fig. 7b**). Additionally, all the three discharge curves in **Fig. 7b** have obvious IR drop. The IR drop of 10% $MnFe_2O_4$@PANIelectrode is less than $MnFe_2O_4$ and PANI which demonstrates that the 10% $MnFe_2O_4$@PANI nanocomposites have smaller internal resistance.

Also, the charge-discharge measurements of 10% $MnFe_2O_4$@PANI nanoflower electrodes were studied in the full cell at various current densities to know the rate performance of the electrode **(Fig. 7c**). The observed drop in the capacitance with increase in the current density in **Fig. 7c** is attributed to reduced electrochemical utilization of the electroactive materials as well as low diffusion of electrolytic ions at high current density. In addition, at the beginning of the discharge curves, there are sudden IR drop (voltage drop) owing to the internal resistance of diffusion of ion and electrolyte resistance [50].

Additionally, with decrease in the current density, the charge-discharge time increases significantly which suggests enhanced diffusion resistance of the electrolytic ions in the 10% $MnFe_2O_4$@PANI nanocomposite electrode material.

3.4.2. Cycling performances

Furthermore, to understand the practical application of the nanocomposites, chronopotentiometric galvanostatic charge-discharge (GCD) was performed to examine the cycling performance and the electrochemical capacitance of the 10%$MnFe_2O_4$@PANI based electrode (**Fig. S6). Fig. 7d** represents the durability and stability of the 10% $MnFe_2O_4$@PANI nanocomposite electrode material with repeated charge-discharge measurement at constant current density of 1 A/g. As observed from **Fig. 7d**, the IR drop increases after 10,000 charge-discharge cycles which is due to the increase in the internal resistance which is explained well from the Nyquist plot of 10% $MnFe_2O_4$@PANI nanocomposite electrode material before and after cycling. The 10%$MnFe_2O_4$@PANI electrode showed a specific capacitance of 623 F/g using **equation (2**) with high cyclic stability upto 10,000 cycles. In addition, the specific capacity of the device was calculated to be ~878 mAh/g. This could be attributed to the fluffy nanoflower morphology of polyaniline, which increased the surface area for better penetration of electrolytic ions into the surface of the electrode. The doping of PANI and the spinel nano-ferrites helped in storing the charge, thereby enhancing the capacitance of the electrode. **Fig. 7a** represents the Ragone plot of energy density *versus* power density of 10%$MnFe_2O_4$@PANI [51–58].

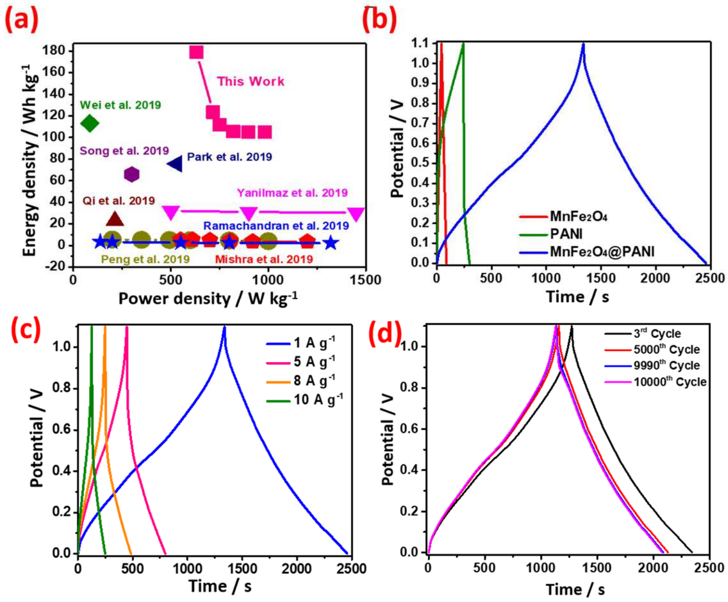


Fig. 7 (a) Ragone plot for the 10%$MnFe_2O_4$@PANI nanoflower, (b) GCD curves of $MnFe_2O_4$, PANI and $MnFe_2O_4$@PANI nanoflower electrodes at a current density of 1 A/g, (c) GCD curves of 10%$MnFe_2O_4$@PANI electrode recorded at different current densities, and (d) Charge-discharge behaviour of symmetric device at 3rd, 5000th, 9990th and 10000th cycles.

The specific capacitance (C_s), energy density (E) and power density (P) for the supercapacitor were calculated using the following equations:

$$C_s = \frac{2I\Delta t}{m\Delta v} \qquad (2)$$

$$E = \frac{C_s \Delta V^2}{2 \ x \ 3.6} \qquad (3)$$

$$P = \frac{3600E}{\Delta t} \qquad (4)$$

The energy densities of the electrodes were calculated to be 179, 123, 111, 105, 104, and 104 $Whkg^{-1}$ at power densities of 631, 716, 749, 820, 898, and 982 Wkg^{-1}, respectively for the exemplar cell with the mass loading of 4.8 Wh/cm^2.

Electrochemical impedance spectroscopy (EIS) was used to examine the internal resistance of the electrode and the resistance at the electrode-electrolyte interface. The frequency response of PANI, $MnFe_2O_4$ and 10%$MnFe_2O_4$@PANI nanoflower in the range of 100 kHz to 0.01 Hz at an AC amplitude of 50 mV is shown by Nyquist plot (**Fig. 5d**). From the Nyquist plots, the intersecting points at high-frequency region with real axis are ascribed to the internal resistance (R_s) which includes the contact resistance at the interface of current collector and electrode material, inherent resistance of the electrode material and the resistance of the electrolyte solution. A semicircle was obtained at the high-frequency region, which signified the charge transfer resistance (R_{ct}) occurring at the interface of electrode-electrolyte, bulk solution resistance (R_s). The CPE represented the constant phase element which is not pure capacitance. From **Fig. 5d**, we can observe three semicircles, two of which are very clear in the nanoflower composites and PANI, while the semicircle for the nanoparticles is located in the high-frequency range.

The charge transfer resistance values were found to be approximately 286, 57.4, and 2.8 kΩ for pure PANI, $MnFe_2O_4$, and 10% $MnFe_2O_4$@PANI nanoflowers, respectively. As can be observed from the R_{ct} values, the nanocomposite had the least charge transfer resistance. This suggests that 10% $MnFe_2O_4$@PANI display brisk adsorption of electrolyte on the electrode's surface. Further, the electrochemical behaviour of 10% $MnFe_2O_4$@PANI electrode, before and after 10,000 charge-discharge cycles were evaluated by EIS (**Fig. S13b**). From the impedance data, equivalent circuit model have been predicted. The charge transfer resistance values for the 10% $MnFe_2O_4$@PANI electrode was found to be ~3.7 kΩ. As can be seen, the R_{ct} value for 10% $MnFe_2O_4$@PANI electrode after 10,000 charge-discharge cycles increases from ~2.8 kΩ to ~3.7 kΩ. This result can be explained according to the FE-SEM image of the 10% $MnFe_2O_4$@PANI nanocomposite electrode after 10,000 charge-discharge cycles (Fig. S8). The mechanical degradation of PANI chains in the nanocomposite during high charge-discharge cycles reduces the porosity of electrode material and hence, the charge transfer resistance increases substantially, due to challenging migration of the electrolyte ions into their interior pores. Inspite of the alleviated mechanical degradation caused by the $MnFe_2O_4$ nanoferrites, there were no sharp enhancement in R_{ct} value of the nanocomposite electrode [59].
Bode plot of impedance *versus* frequency and phase angle *versus* frequency is represented in **Fig. 5e** and **Fig. 5f**, respectively. **Fig. 5e** shows decrease in the value of impedance as frequency increases, which displays capacitor behavior [60]. Additionally, the time constant and frequency response of the electrode material were observed from the Bode plot of phase angle against frequency. At 45˚ phase angle for $MnFe_2O_4$, PANI and $MnFe_2O_4$@PANI gave frequency response (f_O) at 0.604, 0.860 and 2.079 Hz, respectively. The response of relaxation time constant (τ_O) were obtained using the following equation;

$$\tau_O = 1/f_O$$

From the 45˚ phase angle frequency, the relaxation time constant values were evaluated to be 1.65, 1.16 and 0.48 s for $MnFe_2O_4$, PANI and $MnFe_2O_4$@PANI, respectively. The $MnFe_2O_4$@PANI nanocomposite electrode took very less time compared to the other two which is mostly due to its low resistance value.
Table S3 displays the detail information obtained from XPS of 10%$MnFe_2O_4$@PANI composite electrode after 10,000 GCD cycles regarding the shift in energy in each bond, based on which the mechanism has been interpreted. **Fig. S7a** illustrates the wide scan spectra and **Fig. S7b**, **S7c** and **S7d** illustrate the core level XPS spectra *C1s, N1s,* and *O1s* respectively of 10%$MnFe_2O_4$@PANI electrode after 10,000 GCD cycles. The binding energy of O*1s* electrons, is a measure of the extent to which electrons are confined on oxygen or in the internuclear region, a direct significance of the nature of bonding between

different cations and the oxygen. As can be seen from **Table S4**, the increase in the lower binding energy (~531 eV) of nanocomposite electrode after 10,000 charge-discharge cycles is attributed to the cations forming covalent bond with oxygen whereas, an increase in the higher binding energy (~533 eV) correspond to the ionic or covalent bond with two atoms [61]. From the **Table S4**, the increase in FWHM of the former peak (1.65) is due to increase in Mn-O/Fe-O oxygens (oxygen in Fe-O-Mn) after performing proton intercalation-deintercalation in presence of acidic electrolyte. This gave a higher binding energy compared to the oxygen in nanocomposite before charge-discharge cycling due to the difference in the electronegativities of the electrolyte ions and the cations of the nanocomposite. However, the area percentage of the later peak has increased to ~80% which is due to increasing concentration of interaction between the electrolyte ions and the nanocomposite electrode. This is in agreement with the proposed mechanism. Further, from **Table S5** the area percentage is 19.3% for the nanocomposite electrode material after 10,000 charge-discharge cycles which is assigned to oxidized amine which is slightly less than the area percentage (19.4%) obtained before charge-discharge cycles. Moreover, increase in the area percentage of the protonated amine (28.2%) obtained for the nanocomposite electrode after 10,000 charge-discharge cycles as compared to the former (17.5%), explains the interaction of the electrolytic ions *viz.* protons with the nanocomposite during charge-discharge leading to a higher percentage of protonated amine; and so shift in FWHM is obtained indicating change in their chemical state, the interaction of N^+ with the protons of the acidic electrolyte. We report the fabrication of a high-performance symmetrical supercapacitor using low-cost metal ferrites and polyaniline for use as an electrode material to commercialize it in large scale.

4. Conclusion

In this study, we have developed a fluffy $MnFe_2O_4$@PANI nanoflower composite through a well-designed synthetic process using a novel greener process. The materials involved and the synthesis route are cheap, environmentally friendly, with negligible use of organic solvents; this cut down the overall production cost and hence can be used as an electrode material to commercialize it in large scale. Furthermore, the ferrite to polyaniline ratio was found to be crucial in governing the electrode's electrochemical performance for use as electrode material in supercapacitor. The possible interaction mechanism of $MnFe_2O_4$@PANI is investigated from XPS. The nanoflower morphology augment the penetration of the electrolytic ion into the composite which significantly enhanced the specific capacitance (623 F/g at 1 A/g) and established good cyclic stability.

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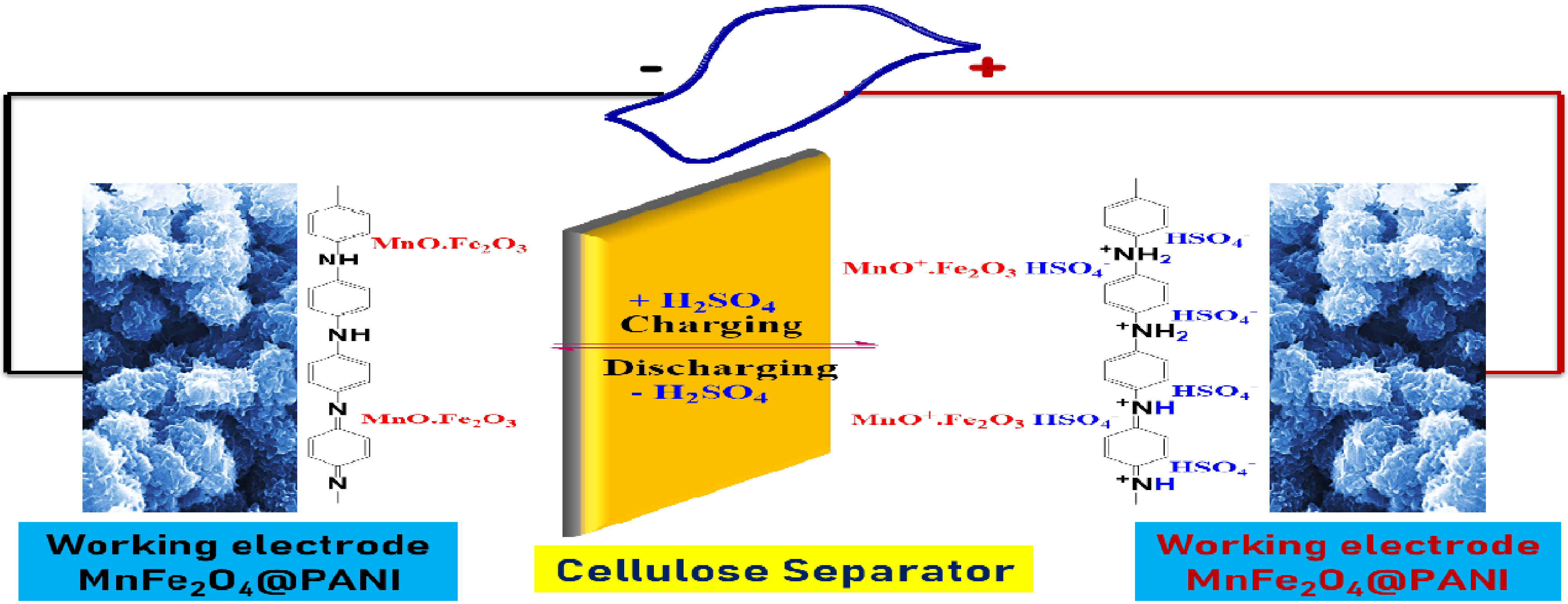

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HSO_4^-
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HSO_4^-
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Working electrode
$MnFe_2O_4$@PANI
Cellulose Separator
Working electrode
$MnFe_2O_4$@PANI